

DAY 03 — TCP/UDP Enumeration & Packet Flow Analysis

Assessment Information

Field	Details
Organization	Trios Cyber
Assessment	Day 03 Assessment
Task	TCP/UDP Enumeration
Date	03 September 2026
Platform	Kali Linux
Target	Metasploitable
Target IP	192.168.56.101
Kali Linux IP	192.168.56.102
Lab Interface	eth0
Tools Used	Nmap, Wireshark, curl

1. Executive Summary

This assessment focused on targeted TCP enumeration, a small authorized UDP scan, identification of network services, and packet-flow analysis using Wireshark.

The assessment was performed against the intentionally vulnerable Metasploitable virtual machine within an isolated and authorized local VirtualBox laboratory environment.

The TCP assessment identified multiple open services on selected ports, while the UDP assessment identified several accessible UDP services. A relevant HTTP/TCP packet flow between Kali Linux and the Metasploitable target was also captured and examined using Wireshark.

No exploitation or unauthorized activity was performed during this assessment.

2. Assessment Objectives

The objectives of Day 03 were:

1. Perform targeted TCP port enumeration using Nmap.
2. Perform a small authorized UDP scan against selected ports.
3. Identify at least three services exposed by the target.
4. Capture one relevant packet flow using Wireshark.
5. Analyze the relationship between the network traffic and the identified service.
6. Preserve command output and screenshots as assessment evidence.
7. Document the methodology, findings, and learning outcomes professionally.

3. Authorization and Scope

All testing was conducted within the authorized local cybersecurity laboratory environment.

Assessment Scope

- **Attacker/Assessment System:** Kali Linux
- **Assessment IP:** 192.168.56.102
- **Target System:** Metasploitable

- **Target IP:** 192.168.56.101
- **Network:** VirtualBox Host-Only Lab
- **Interface:** eth0
- **Testing Type:** TCP/UDP enumeration and packet capture
- **Authorization:** Local laboratory environment only

The assessment was limited to service enumeration and network traffic observation. No exploitation, credential attacks, denial-of-service activity, or testing against public systems was performed.

4. Lab Environment

The assessment environment consisted of two virtual machines connected through a VirtualBox Host-Only network.

Kali Linux

- IP Address: 192.168.56.102/24
- Interface: eth0
- Purpose: Assessment and enumeration system

Metasploitable

- IP Address: 192.168.56.101
- Purpose: Intentionally vulnerable laboratory target

The Host-Only network allowed Kali Linux to communicate directly with the Metasploitable target without exposing the assessment traffic to an external network.

5. Network Connectivity Verification

Before performing enumeration, connectivity between Kali Linux and the Metasploitable target was verified.

Command

```
ping -c 4 192.168.56.101
```

Evidence: screenshots/01-network-connectivity.png

6. Targeted TCP Enumeration

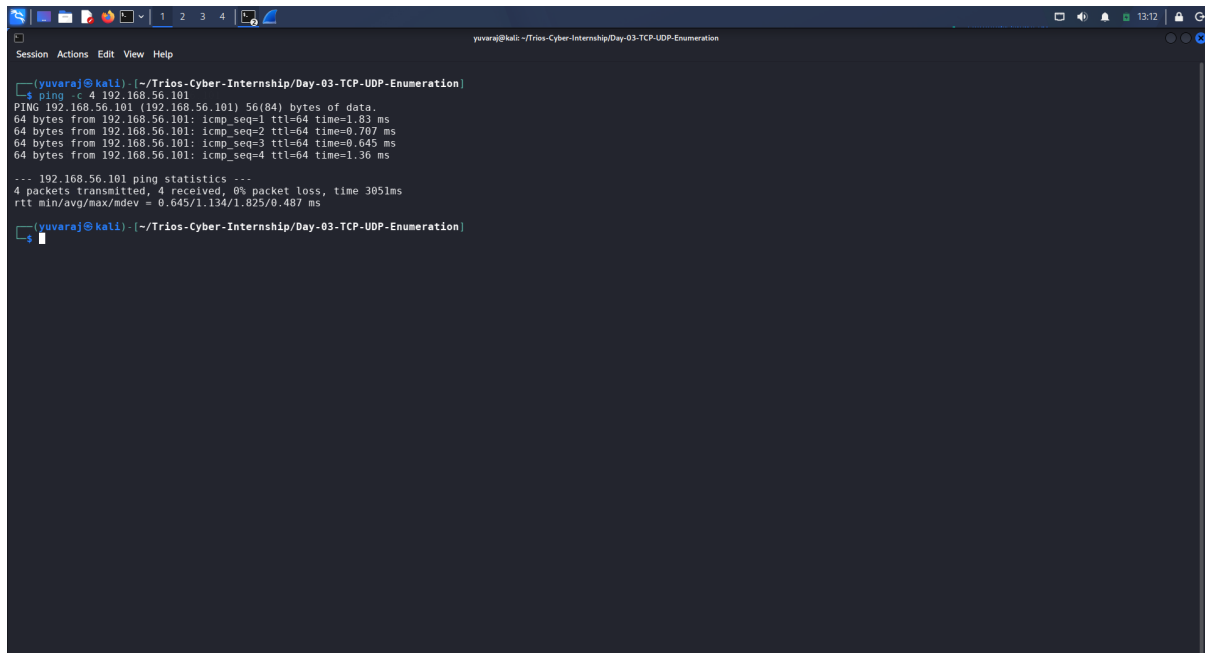
A targeted TCP scan was performed against selected ports commonly associated with network and application services.

Command

```
nmap -sT -p 21,22,23,25,53,80,139,445,3306,5432,5900,8009,8180 192.168.56.101 | tee logs/tcp-targeted
```

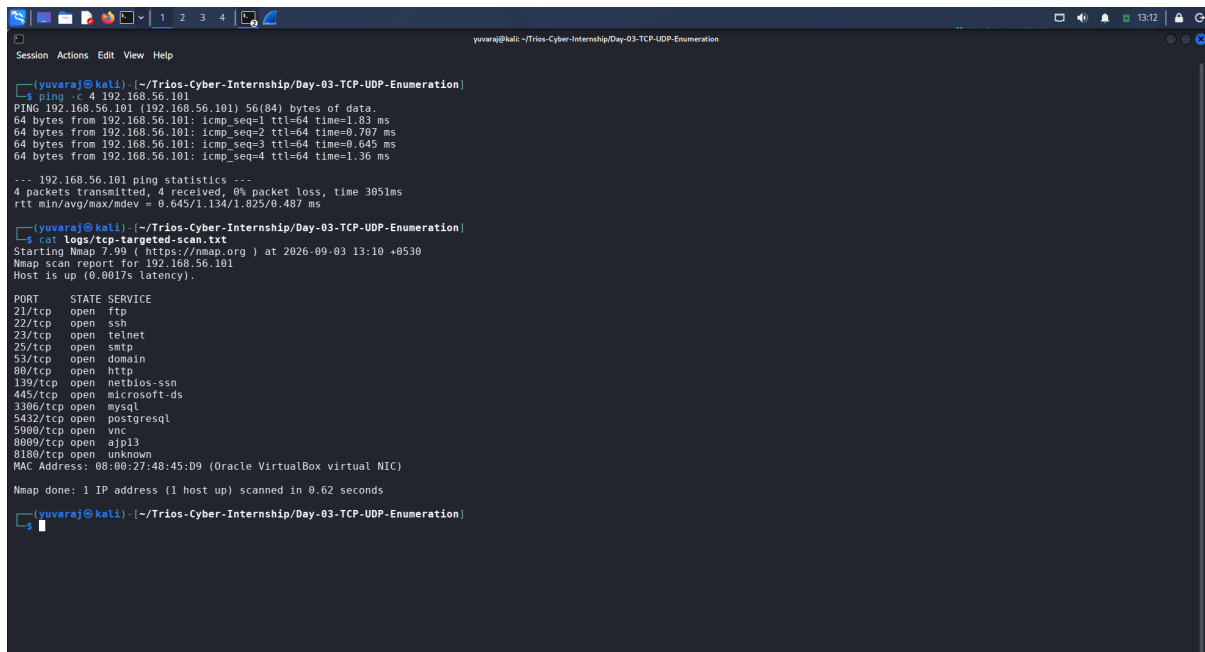
Evidence: screenshots/02-tcp-enumeration.png

Raw Output: logs/tcp-targeted-scan.txt



```
yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.83 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.707 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.645 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.36 ms
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.645/1.134/1.825/0.487 ms
yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
```

Figure 1: Network connectivity verification



```
yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
$ cat logs/tcp-targeted-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:10 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0017s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

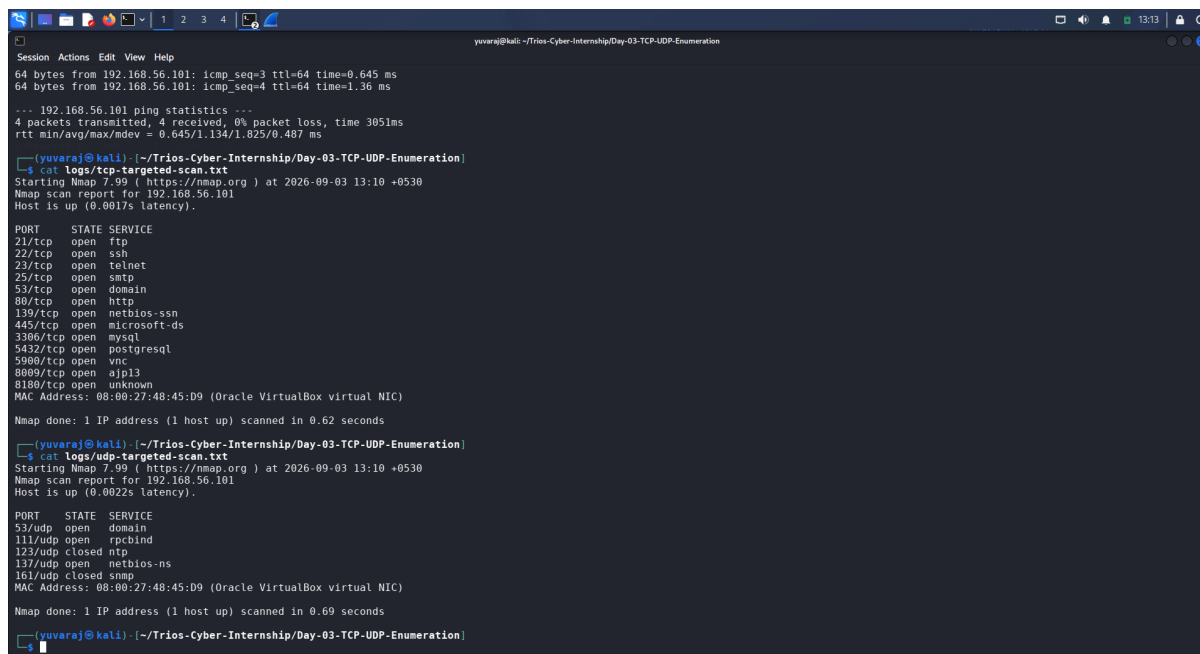
Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
```

Figure 2: Targeted TCP enumeration results

7. Small UDP Enumeration

A small targeted UDP scan was performed against five selected UDP ports. The scan was intentionally limited to the authorized Metasploitable laboratory target.

```
nmap -sU -p 53,111,123,137,161 192.168.56.101 | tee logs/udp-targeted-scan.txt
```



```
Session Actions Edit View Help
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.645 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.36 ms

... 192.168.56.101 ping statistics ...
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.645/1.134/1.825/0.487 ms

[yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration]
$ cat logs/tcp-targeted-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:10 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0017s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds

[yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration]
$ cat logs/udp-targeted-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:10 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0022s latency).

PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
123/udp   closed ntp
137/udp   open  netbios-ns
161/udp   closed snmp
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds

[yuvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration]
$
```

Figure 3: Targeted UDP enumeration results

Evidence: screenshots/03-udp-enumeration.png

Raw Output: logs/udp-targeted-scan.txt

8. Services Identified

The enumeration phase successfully identified multiple services exposed by the Metasploitable target.

TCP Services

The following TCP services were identified:

- FTP — TCP/21
- SSH — TCP/22
- Telnet — TCP/23
- SMTP — TCP/25
- DNS/Domain — TCP/53
- HTTP — TCP/80
- NetBIOS-SSN — TCP/139
- Microsoft-DS — TCP/445
- MySQL — TCP/3306
- PostgreSQL — TCP/5432
- VNC — TCP/5900
- AJP13 — TCP/8009

TCP/8180 was reported as open, but Nmap recorded the service as `unknown` in the captured output. Therefore, no unsupported service identification is assigned to this port.

UDP Services

The following UDP services were identified as open:

- DNS/Domain — UDP/53
- RPCBind — UDP/111
- NetBIOS Name Service — UDP/137

The results demonstrate that the target exposes services over both TCP and UDP.

9. Wireshark Packet Flow Capture

To demonstrate an actual application-layer network flow, HTTP traffic between Kali Linux and the Metasploitable web service was generated and captured using Wireshark.

Traffic Generation

The following command was used to generate an HTTP request:

```
curl -I http://192.168.56.101
```

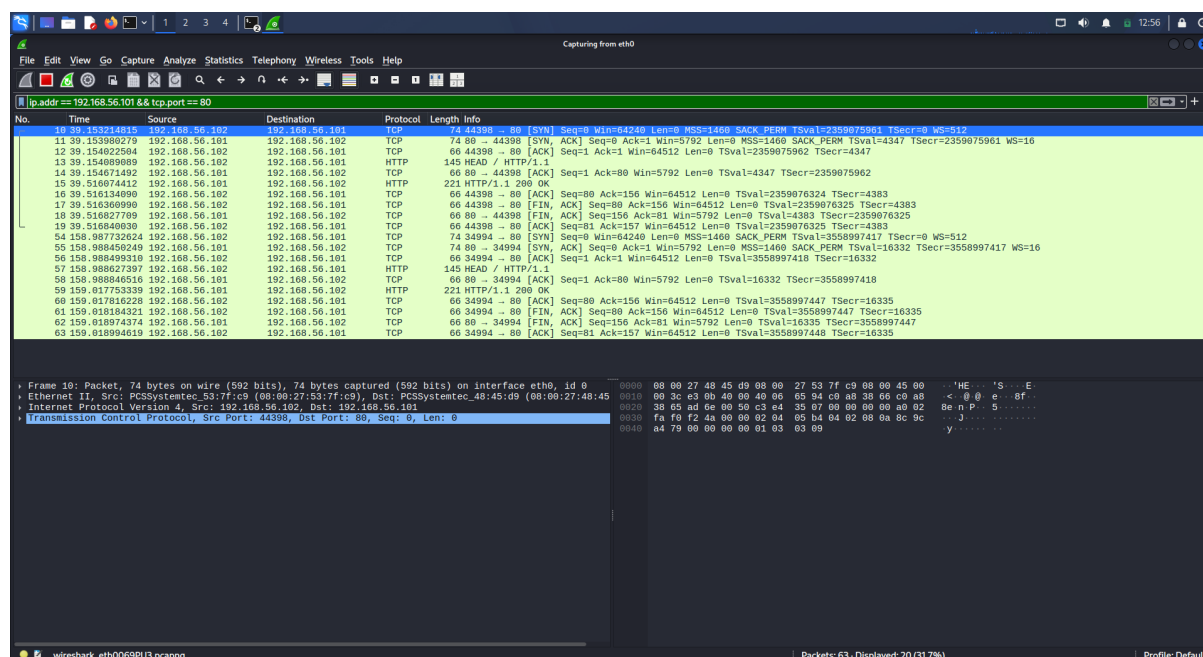


Figure 4: Wireshark HTTP TCP packet flow

Evidence: screenshots/04-wireshark-http-flow.png

10. Network Configuration Evidence

The network configuration was reviewed to confirm the assessment interface and laboratory addressing.

The Kali Linux Host-Only interface used for communication with the Metasploitable target was eth0.

The assessment network used the 192.168.56.0/24 range, with Kali Linux assigned 192.168.56.102 and the Metasploitable target assigned 192.168.56.101.

Evidence: screenshots/05-network-configuration.png

```

yuvvaraj@kali: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
Se  Text Editor
Simple TextEditor  Help
00xvi  STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
23/tcp  open  telnet
25/tcp  open  smtp
53/tcp  open  domain
80/tcp  open  http
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds

[yuvvaraj@kali]: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
$ cat logs/udp-targeted-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-03 13:10 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0022s latency).

PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
123/udp   closed ntp
137/udp   open  netbios-ns
161/udp   closed snmp
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds

[yuvvaraj@kali]: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
$ ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:7f:c9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever

[yuvvaraj@kali]: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
$ ip route
default via 10.0.3.2 dev eth1 proto dhcp src 10.0.3.15 metric 100
10.0.3.0/24 dev eth1 proto kernel scope link src 10.0.3.15 metric 100
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.102 metric 101

[yuvvaraj@kali]: ~/Trios-Cyber-Internship/Day-03-TCP-UDP-Enumeration
$

```

Figure 5: Kali network configuration

11. Security Observations

The enumeration results provide several observations relevant to network security.

11.1 Multiple Exposed Services

The target exposes numerous TCP services, including FTP, Telnet, HTTP, SMB-related services, database services, and VNC.

A larger number of exposed services increases the potential attack surface of a system and should be reviewed during a security assessment.

11.2 Legacy and Remote Access Services

Services such as Telnet and FTP are examples of network services that require careful security evaluation because their presence can introduce additional security concerns depending on configuration and authentication controls.

11.3 Database Services

MySQL and PostgreSQL were identified on the selected TCP ports. Database services should generally be restricted to trusted hosts and protected using appropriate authentication and network-access controls.

11.4 UDP Exposure

DNS, RPCBind, and NetBIOS Name Service were identified as open on the selected UDP ports. UDP services should also be assessed because they form part of the overall network attack surface.

These observations are limited to enumeration results. No vulnerability exploitation was performed as part of this assessment.

12. Assessment Methodology

The assessment followed a controlled sequence:

1. Confirm the authorized target and assessment scope.
2. Verify network connectivity between Kali Linux and Metasploitable.
3. Perform targeted TCP enumeration.
4. Record the identified open TCP services.
5. Perform a small targeted UDP enumeration.
6. Record the UDP service states.
7. Generate HTTP traffic toward the identified web service.
8. Capture and filter the HTTP/TCP traffic using Wireshark.
9. Preserve raw scan outputs and screenshots.
10. Document the assessment results and observations.

This methodology provided a controlled demonstration of basic network service enumeration and packet-flow analysis.

13. Evidence Index

The following evidence was collected during the assessment.

Evidence	Description
logs/assessment-scope.txt	Assessment scope and target information
logs/tcp-targeted-scan.txt	Targeted TCP Nmap scan output
logs/udp-targeted-scan.txt	Targeted UDP Nmap scan output
screenshots/01-network-connectivity.png	Successful connectivity verification
screenshots/02-tcp-enumeration.png	Targeted TCP enumeration results
screenshots/03-udp-enumeration.png	Targeted UDP enumeration results
screenshots/04-wireshark-http-flow.png	HTTP/TCP packet-flow capture
screenshots/05-network-configuration.png	Kali network configuration

All evidence was collected from the authorized local laboratory environment.

14. Key Findings Summary

Category	Finding
Connectivity	Kali Linux successfully communicated with the target
TCP Enumeration	13 selected TCP ports were reported open
UDP Enumeration	3 selected UDP ports were reported open
Services	Multiple network and application services were identified
HTTP	TCP/80 was identified as open
Packet Capture	HTTP/TCP traffic was successfully observed in Wireshark
Scope	Testing remained within the authorized local lab

15. Learning Outcomes

This assessment provided practical experience with:

- Targeted TCP port enumeration using Nmap.
 - Basic UDP service enumeration.
 - Understanding TCP and UDP service exposure.
 - Identifying services from Nmap scan results.
 - Using Wireshark to observe network packets.
 - Applying Wireshark display filters.
 - Correlating an identified service with observed network traffic.
 - Preserving raw technical evidence.
 - Documenting cybersecurity assessment activities professionally.
-

16. Conclusion

Day 03 successfully demonstrated targeted TCP and UDP enumeration against the authorized Metasploitable laboratory system.

The TCP scan identified 13 open ports among the selected targets, including FTP, SSH, Telnet, HTTP, MySQL, PostgreSQL, VNC, and other services. The targeted UDP scan identified DNS, RPCBind, and NetBIOS Name Service as open among the selected UDP ports.

An HTTP/TCP packet flow was also generated and captured using Wireshark, providing practical evidence of communication with the web service identified during enumeration.

The assessment was conducted exclusively within the authorized local VirtualBox laboratory environment. No exploitation or unauthorized testing was performed.

17. Appendix — Commands Executed

Connectivity Verification

```
ping -c 4 192.168.56.101
```

Targeted TCP Enumeration

```
nmap -sT -p 21,22,23,25,53,80,139,445,3306,5432,5900,8009,8180 192.168.56.101
```

Targeted UDP Enumeration

```
nmap -sU -p 53,111,123,137,161 192.168.56.101
```

HTTP Traffic Generation

```
curl -I http://192.168.56.101
```

Wireshark Display Filter

```
ip.addr == 192.168.56.101 && tcp.port == 80
```

Assessment Completion Status

Assessment Status: Completed

Scope Status: Authorized Local Laboratory Only

Evidence Status: Complete

Report Status: Ready for PDF Generation and Git Submission